

COMP5310 – Sample Questions

In Preparation for the Final Exam

Overview

This document aims to help you prepare for your COMP5310 Semester 2, 2025 final exam by providing you with (1) key exam information and (2) practice questions.

Key Exam Information

- **Date and time:** Saturday, November 22, 2025, at 9:00 AM.
- **Weighting:** 60% of the overall unit mark.
- **Structure:** The exam paper comprises 7 sections. Mark's total is 100. Each section contains multiple sub-questions.
 - Question 1 (7 sub-questions, 10 marks)
 - Question 2 (3 sub-questions, 15 marks)
 - Question 3 (4 sub-questions, 15 marks)
 - Question 4 (11 sub-questions, 15 marks)
 - Question 5 (5 sub-questions, 15 marks)
 - Question 6 (3 sub-questions, 15 marks)
 - Question 7 (3 sub-questions, 15 marks)
- **Time:** 2 hours writing time, plus 10 minutes reading time.
- **Exam condition:** This is a RESTRICTED OPEN book exam - specified materials permitted.
- **Materials permitted:** (1) Calculator – non-programmable, (2) One A4 sheet of handwritten and/or typed notes, double-sided, (3) Students may also use a linguistic dictionary.
- **Key instructions:**
 - Answer in the exam paper itself and within the space provided.
 - ANSWER ALL QUESTIONS using a blue or black pen on this examination booklet. Take care to write legibly.
 - Please check that your examination paper is complete.
 - This question paper must be returned with any answer booklets or materials (as the A4 sheet of your own notes)

Sample Question 1: Multiple Choice Questions

[Q1.1] What is the main difference between the interval datatype and the ratio datatype?

- A. Interval data have an absolute zero point, while ratio data do not.
- B. Ratio data have an absolute zero point, while interval data do not.
- C. Both interval and ratio data have absolute zero points.
- D. Ratio data cannot be used for arithmetic operations, while interval data can.
- E. Interval data can only represent categories without order.

[Q1.2] Is it meaningful to order nominal data?

- A. Yes, because nominal data have a natural ranking.
- B. Yes, but only if the categories are numeric.
- C. No, because nominal data represent categories without any inherent order.
- D. No, because nominal data are always continuous.

[Q1.3] Does the median measure the central tendency or dispersion?

- A. Central tendency
- B. Dispersion
- C. Both central tendency and dispersion
- D. Neither

[Q1.4] Which of the following regarding the k-means algorithm is/are correct?

- A. The k-means algorithm takes k as an input.
- B. The k-means algorithm automatically computes the appropriate value of k.
- C. The k-means algorithm minimizes within-cluster variance.
- D. K-means assigns each data point to the cluster with the nearest centroid.

Sample Question 2: Association Rules

Given the following transaction record

Transaction ID	Items
#1	apple, banana, coca-cola, doughnut
#2	banana, coca-cola
#3	banana, doughnut
#4	apple, coca-cola
#5	apple, banana, doughnut
#6	apple, banana, coca-cola

[Q2.1] Build the FP-tree using a minimum support $\text{min_sup} = 2$. Show how the tree evolves for each transaction.

[Q2.2] With the previous transaction record, use the Apriori algorithm on this dataset and verify that it will generate the same set of frequent itemsets with $\text{min_sup} = 2$.

[Q2.3] Suppose that $\{\text{Apple, Banana, Doughnut}\}$ is a frequent item set, derive all its association rules with $\text{min_confidence} = 70\%$

Sample Question 3: Database Systems

[Q3.1] You have two tables, employees and departments, in your database. The employees table contains the following columns: employee_id, first_name, last_name, department_id, and salary. The departments table contains the columns: department_id and department_name. Write a SQL query to retrieve the first_name of each employee along with the department_name they belong to.

[Q3.2] Assume you have four datasets: "employees", "departments", "projects", and "assignments". The scheme of these databases is as follows:

- The "employees" dataset contains employee information, including a unique "employee_ID", "employee_name", and "department_ID".
- The "departments" dataset contains department information, including a unique "department_ID" and "department_name".
- The "projects" dataset contains project information, including a unique "project_ID", "project_name", and "department_ID".
- The "assignments" dataset contains information about which employees are assigned to which projects, including the "employee_ID" and "project_ID" associated with each assignment.

Write a SQL query to retrieve the name and department of all employees who are assigned to a project that is not in the same department as their own department.

Sample Question 4: Naïve Bayes Classifier

A national park has created a dataset to help hikers determine if a reptile they encounter could be venomous.

	Head	Eyes	Size	Venomous
1	Triangle	Elliptical	Small	Yes
2	Round	Round	Small	No
3	Narrow	Elliptical	Small	No
4	Narrow	Round	Large	No
5	Narrow	Elliptical	Large	Yes
6	Triangle	Round	Small	Yes
7	Narrow	Round	Large	No
8	Round	Elliptical	Large	No
9	Triangle	Elliptical	Small	Yes

Use Naïve Bayes to predict if the following example is venomous or not:

Head=narrow, Eyes=elliptical, Size=Large

Show the working for your calculations.

Sample Question 5: Hierarchical Clustering

The following pages trace a hierarchical clustering of distances in miles between U.S. cities. The method of clustering is single-link. Use the hierarchical clustering algorithm to cluster the following data. Illustrate every step of the merging process.

	BOS	NY	DC	MIA	CHI	SEA	SF	LA	DEN
BOS	0	206	429	1504	963	2976	3095	2979	1949
NY	206	0	233	1308	802	2815	2934	2786	1771
DC	429	233	0	1075	671	2684	2799	2631	1616
MIA	1504	1308	1075	0	1329	3273	3053	2687	2037
CHI	963	802	671	1329	0	2013	2142	2054	996
SEA	2976	2815	2684	3273	2013	0	808	1131	1307
SF	3095	2934	2799	3053	2142	808	0	379	1235
LA	2979	2786	2631	2687	2054	1131	379	0	1059
DEN	1949	1771	1616	2037	996	1307	1235	1059	0

Sample Question 6: Classification Evaluations

Imagine you are working on a project that is a binary classification problem. You trained a model on the training dataset and got the following confusion matrix on the validation dataset.

	Predicted: No	Predicted: Yes
Actual: No	100	150
Actual: Yes	50	200

[Q6.1] Calculate the accuracy, precision, recall, and F1-score.

[Q6.2] Which of the above metrics will you use? Explain the reasons.

Sample Question 7: Decision Trees

Consider the following training dataset that we want to build a decision tree to predict whether an applicant will perform well or not in an interview. Suppose that the “level” attribute is chosen to split the data at the root of the decision tree.

ID	Level	Lang	Tweets	PhD	Perform well
1	Senior	Java	No	No	No
2	Senior	Java	Yes	Yes	No
3	Mid	Java	No	No	Yes
4	Junior	Python	No	No	Yes
5	Junior	R	Yes	No	Yes
6	Junior	R	Yes	Yes	No
7	Mid	R	Yes	Yes	Yes
8	Senior	Python	No	No	No
9	Senior	Python	Yes	No	Yes
10	Junior	Python	Yes	No	Yes
11	Senior	Python	Yes	Yes	Yes
12	Mid	Python	No	Yes	Yes
13	Mid	Java	Yes	No	Yes
14	Junior	Python	No	Yes	No

[Q7.1] List the training examples that fall under each child of the root of the decision tree.

[Q7.2] Use information gain to select the best attribute to split the data for the child node (of the root in the decision tree) corresponding to level = Senior. Show your calculation details.